DIGITAL ELECTRONICS LAB (ESC392)

PART B — GENAI-ASSISTED DESIGN ASSIGNMENT 2

Natural Language to Logic Circuit Generator

1. GenAI Model Name

ChatGPT (OpenAI)

2. Prompt

Design a logic circuit for a simple home security alarm system. The alarm should turn ON if either the door is open or the window is open. Let D represent the door sensor, W represent the window sensor, and A represent the alarm output. Convert this English description into a Boolean expression, truth table, and gate-level circuit. Explain how to implement and verify it in Logisim Evolution.

3. Response

The alarm system turns ON when either the door or the window is open.

- D = 1: Door is open.
- W = 1: Window is open.
- A = 1: Alarm is ON.

Boolean Expression:

A = D + W

Here, + represents the OR operation.

Truth Table:

D	W	A
0	0	0
0	1	1
1	0	1
1	1	1

The circuit requires one 2-input OR gate.

4. Verification by Logisim

1. Open Logisim Evolution and create a new circuit.

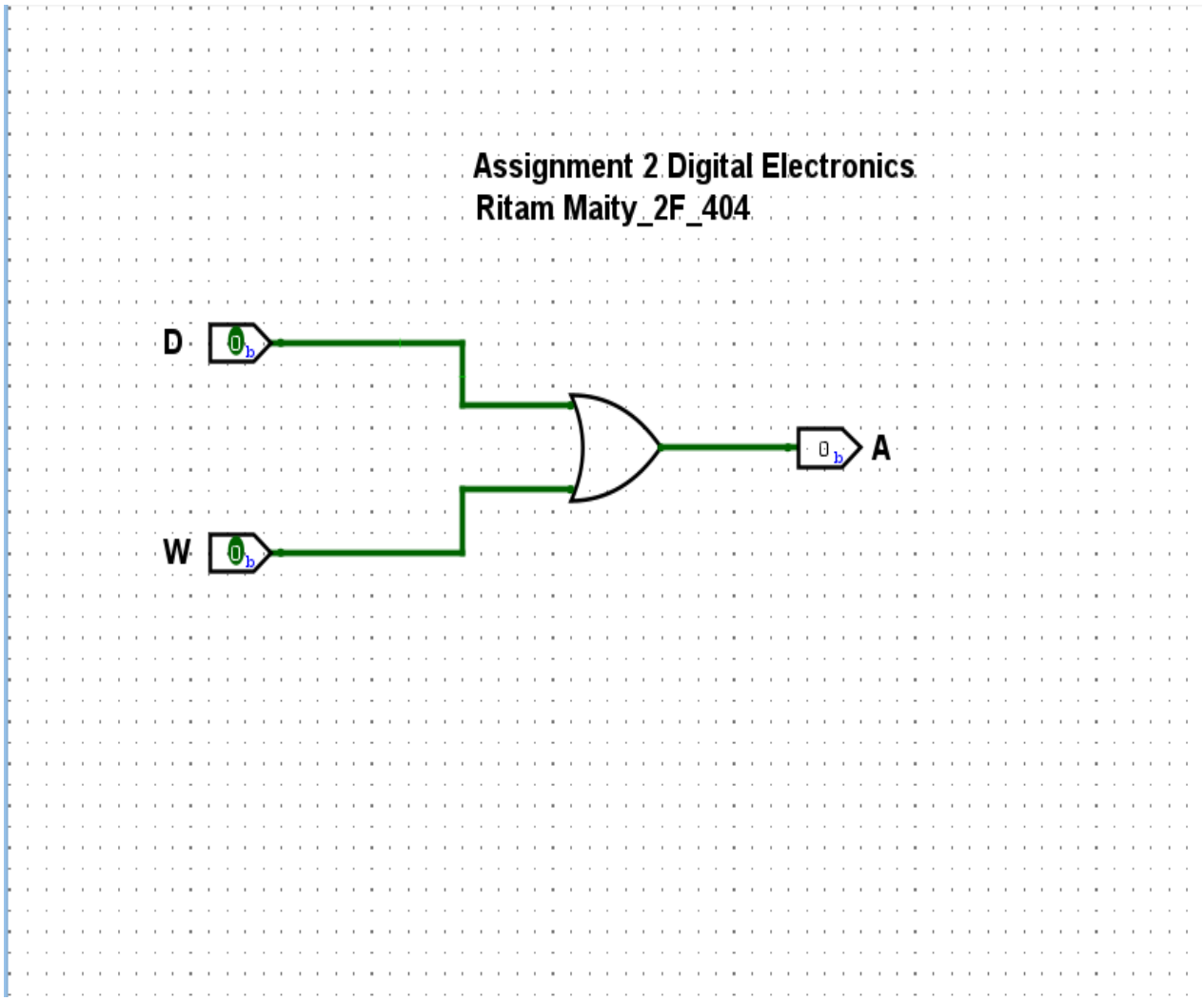
2. Place two input pins and label them D and W.
3. Place one 2-input OR gate.
4. Connect both input pins to the OR gate.
5. Connect the output of the OR gate to an output pin labelled A.
6. Use the Poke Tool to test all four input combinations.
7. Verify that the simulated outputs match the truth table.

5. Circuit:

Circuit connections:

Input D ─┐
OR Gate ── Output A
Input W ─┘

The two inputs D and W are connected to the OR gate, and its output represents the alarm signal A.



6. Output

The expected output is:

D	W	Expected A
0	0	0
0	1	1
1	0	1
1	1	1

The output becomes HIGH if at least one input is HIGH. Insert the actual Logisim simulation screenshot here after testing the circuit.

7. Discussion

The home security alarm circuit was designed using a 2-input OR gate. The alarm turns ON when either the door or the window is open. When both are closed, the alarm remains OFF. The truth table is used to verify the correct operation of the circuit.

8. GitHub Link

GitHub Repository Link: https://github.com/Ritam-Maity/Digital-Electronics-Lab

The report and Logisim circuit file (.circ) should be uploaded to GitHub, and the repository link should be included in the report.